

Cassette 4

A cassette four-track as a summing mixer: four stereo stems recorded to tape and mixed down.



Overview

Cassette 4 is a summing mixer in the spirit of the 80s portable multitrack recorders. Each of four stereo stems goes to its own track of the tape - recorded, played back through the heads - and is then mixed down through the recorder's little mixer with its simple tone controls, pan and level.

Every track gets its own tape path: noise-reduction encoding, tape saturation, the wobble of the transport, the head's top-end loss and bump, dropouts and hiss, and the decoding on the way back. All four tracks share one transport, so they wow and flutter together, just like on a real machine.

Two models are on board. MODEL 15 is the compact standard-speed machine: soft top end, fat head bump, more wow and hiss. MODEL 44 is the double-speed machine with compander noise reduction: more top end and steadier, with the gentle breathing and pumping that compander noise reduction is known for.

Reach for it for

- Lo-fi and bedroom-demo mixes
- Giving a whole track that warm, wobbly cassette character
- Gluing drums, keys and vocals with tape saturation
- Pumping, breathing tape compression (MODEL 44)
- Worn-out, nostalgic textures (AGE, HISS, WOW)

Setting up the routing in REAPER

1. Create a bus track and put Cassette 4 on it. Open the track's ROUTE window and set Track channels to 8.
2. On each stem track, add a send to the bus track and pick its destination channel pair in the send's audio routing: TRACK 1 on 1/2, TRACK 2 on 3/4, TRACK 3 on 5/6, TRACK 4 on 7/8.
3. Turn off the master send on every stem track, so the stems are heard only through Cassette 4.
4. TRACK 1 also receives whatever plays on the bus track itself (channels 1/2), so keep the bus track empty or use it as track 1.
5. The mix leaves on channels 1/2, so leave the bus track's master send on. The green light under each track name lights when signal arrives.

Controls

The mixer: TRACK 1 to TRACK 4

Four channel strips, each with its input channel pair printed under the name.

Signal light	Glows green when audio arrives on the track and it is being heard.
TREBLE	A high shelf at 10 kHz, -10 to +10 dB. Brighten a track that the tape has softened, or dull it further.
BASS	A low shelf at 100 Hz, -10 to +10 dB.
PAN	A balance control: in the centre both sides play at full level; turning it fades the opposite side down.

LEVEL	The track's mixdown level, from OFF (-60 dB) to +6 dB; 0 dB by default.
SOLO	Solo in place; several tracks can be soloed at once.
MUTE	Silences the track. A muted track stays muted even when soloed.

TAPE

DRIVE (coloured cap)	The recording level, -6 to +12 dB. Hotter means more tape saturation. It is level-matched, so it changes the colour rather than the volume.
WOW	Wow and flutter, 0 to 100 (30 by default): a slow wander plus a faster flutter. One transport moves all four tracks together. The 44 model is steadier at the same setting.
HISS	The tape noise, 0 to 100. The middle (50) is the model's own amount. Each track has its own hiss, so four busy tracks mean four times the noise, as on the real thing.
AGE	An older, worn tape, 0 to 100 (10 by default): the top end goes duller and short dropouts creep in, more often the higher you turn it.
MASTER	The final mix level, -24 to +12 dB.
MODEL 15 / 44	Picks the machine. 15: standard speed, B-type noise reduction, a soft top end around 9.5 kHz, a fatter head bump, more wow and hiss. 44: double speed, compander noise reduction, more top end (around 15 kHz), a steadier transport. The panel changes colour with the model.
NR	Noise reduction in or out. Out gives you the raw tape, hiss and all. In MODEL 44 the compander squeezes the signal 2:1 onto the tape and expands it 1:2 coming back; because the expander follows what comes off the tape, it breathes and pumps a little - turn DRIVE up to hear more of it.

Meters

L / R

VU meters showing the left and right mix output.

The header

PRESETS	Opens the list of factory presets. Choosing one sets the plugin's controls for that job; the input and output stages are left as they are. The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
OUTPUT	Folds the input and output stages out below the panel (and back in). The light shows when they are open.

Output stage (OUTPUT fold-out)

SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
BIAS	Shapes the saturation: more bias adds more even (2nd) harmonics, for a warmer, rounder colour.
LO BAND / HI BAND	Where the low and high saturation bands start (40 - 400 Hz and 2 - 16 kHz).
CEILING / THRESH	With CLIP: the level the peaks are softly clipped at. With LIMIT: the threshold of a brickwall limiter that holds the peaks at -0.3 dBFS - pulling it down limits harder and makes the output louder.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER).

Presets

Bedroom Demo (15)	The standard-speed machine at its default settings, NR in.
Hot and Woolly (15)	Recorded hot, with more wow, hiss and age.
No NR, Old Tape (15)	Noise reduction off, lots of hiss, wobble and wear.
Worn-out Cassette (15)	Driven hard on a badly worn tape with heavy wow.
Clean Double Speed (44)	The double-speed machine at its cleanest.
Pumping Compander (44)	6 dB of drive to bring out the compander's breathing.
Squashed Tape (44)	Driven hard for a dense, squashed tape sound.
44 without NR	The double-speed machine with the noise reduction out.
Reset all to defaults	Every control back to its starting value, including the mixer and the output stage.

Tips

- The presets only change the model and the TAPE section; your track levels, EQ, pans, solos and mutes are left alone (except Reset).
- Use the TREBLE knobs on each track to win back a little top end after the tape, rather than turning AGE down to zero.
- Put drums or the busiest material on one track and pads on another: each track saturates on its own, so the mix stays clearer than squashing everything through one tape.
- Every track adds its own hiss even when nothing is sent to it, so MUTE the tracks you are not using to keep the noise floor down.

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